Beyond Physical Digital Twins: Why Logical Digital Twins Are Essential for AI Factory

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Agenda

- What is a Logical Digital Twin
- Why AI Factories need Logical Digital Twins
- Simulation technologies
- NVIDIA + Keysight
- Agentic AI and the future of simulation

From Logical Simulation to a Logical Digital Twin

PLAN



Day 0

DEPLOY



Day 1

OPERATE



Day 2

High Fidelity Infrastructure Simulation for the Entire Lifecycle of the AI Factory

Key Drivers for the AI Factory Logical digital twins

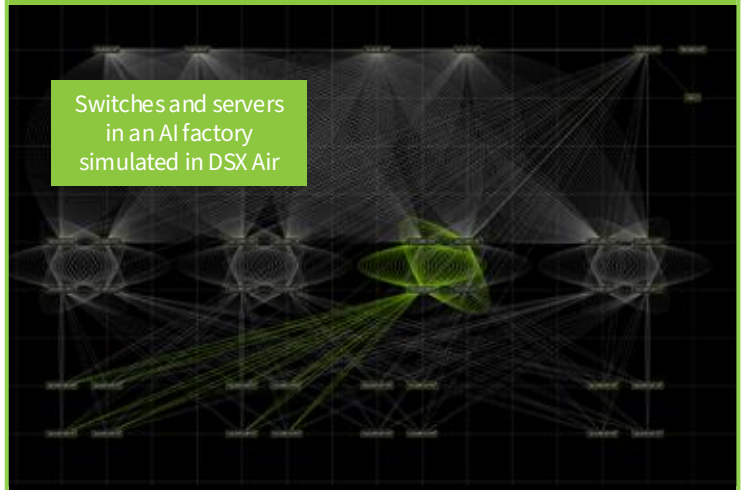
- Every AI Factory needs digital twins for operational simplicity

ECOSYSTEM VALIDATION



Playground to integrate with NVIDIA in scale and partners

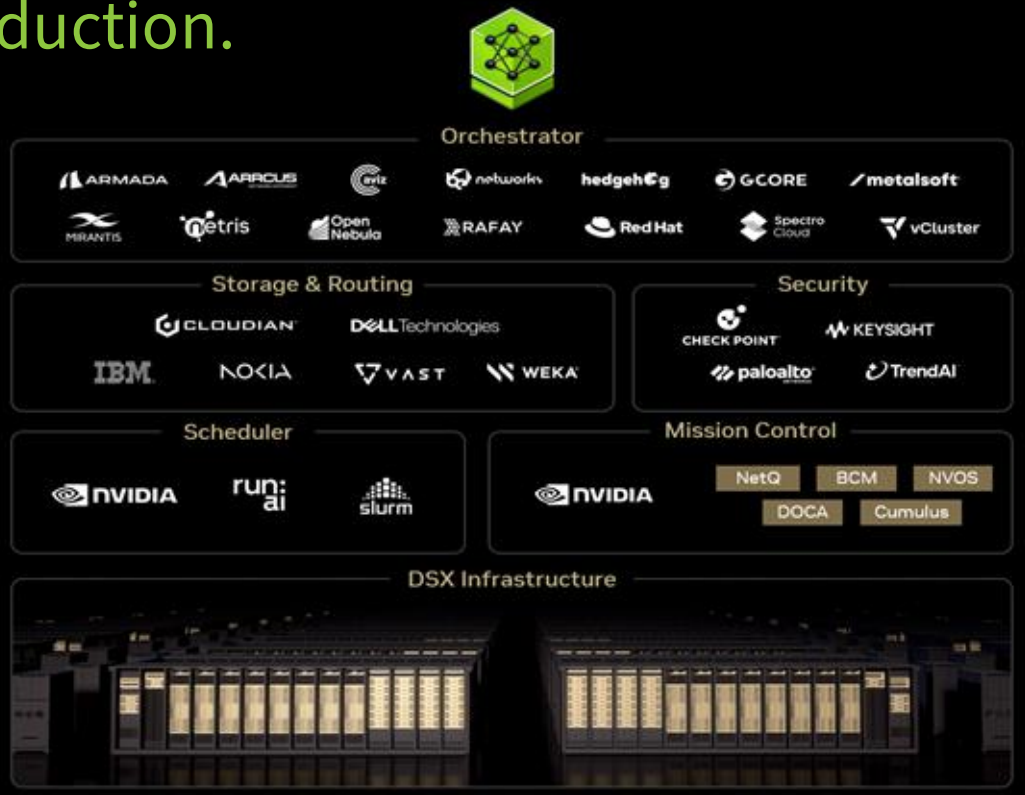
SIMULATION AT SCALE

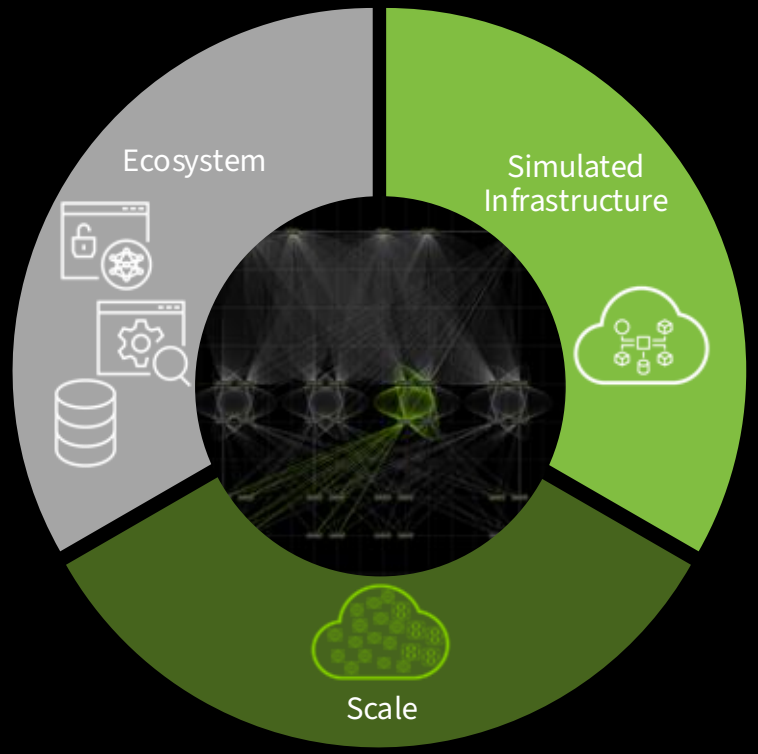


Faster Time to AI

The Logical Twin in the operation cycle

- Every infrastructure change is validated in the Logical Digital Twin before production.





CI/CD Pipeline

Key Challenges a Digital Twin Helps Address

Shift Left



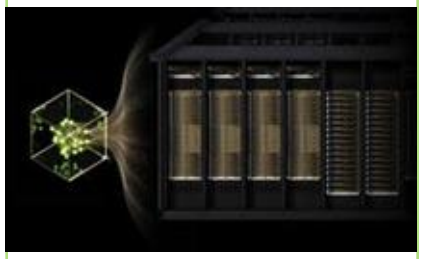
Building your DC before the system Arrival

Democratize AI without compromise



Many Software components
In Agile designs

Sizing & resource constraints



Define the Infrastructure software stack configuration, based on Workload and User requirements

SECURITY



Security aspects both for Users and agents
Alignments between all the components

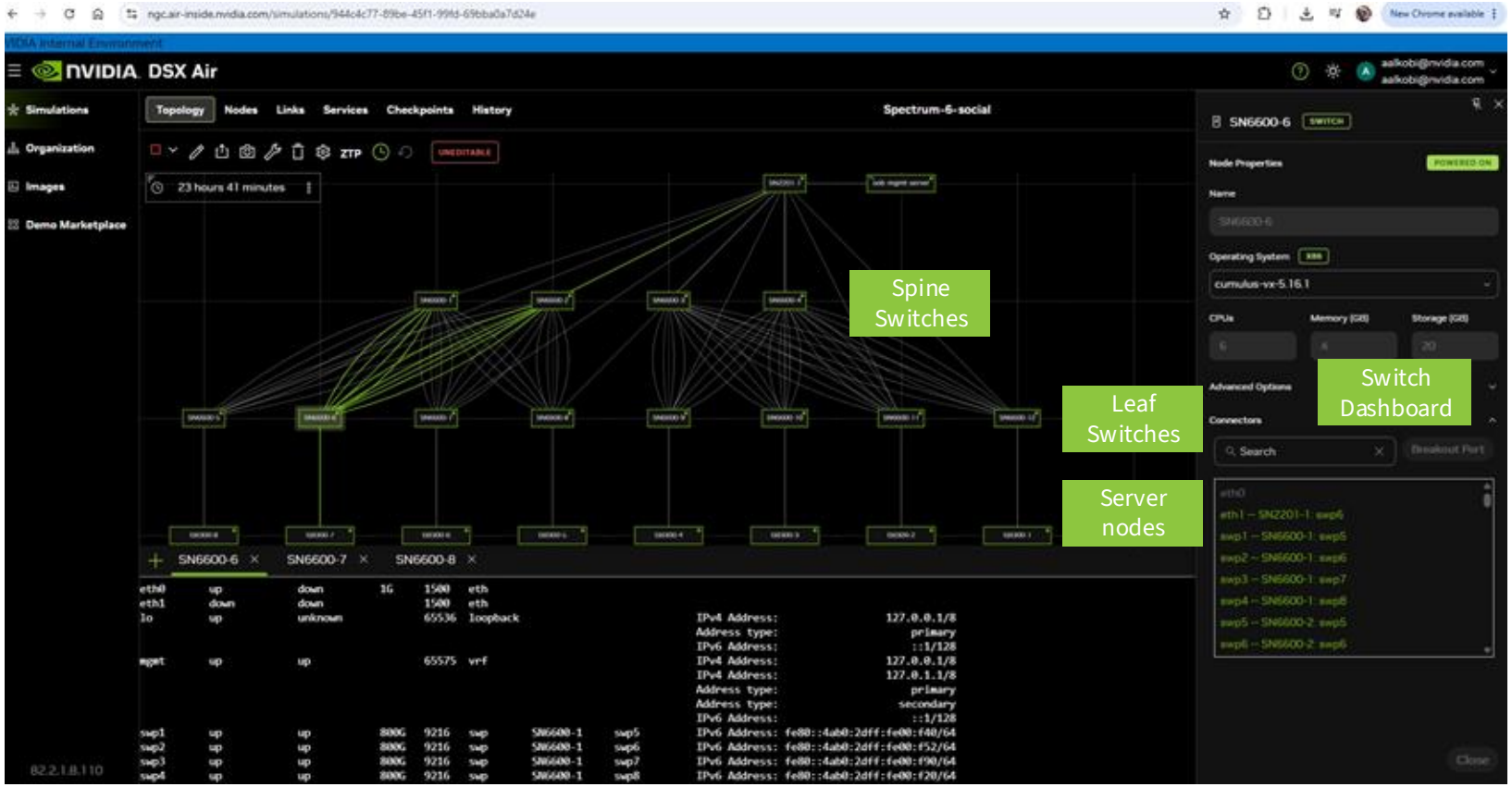
STORAGE



KV Cache considerations

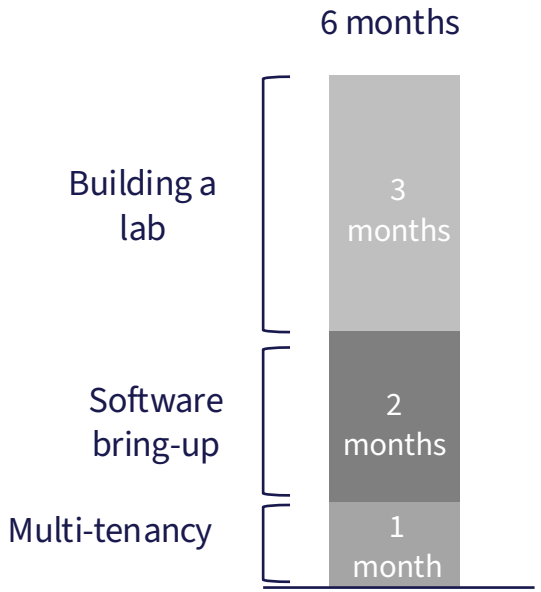
Node based Simulation

- “Shifting left”: accelerating deployment of Spectrum-6 ahead of physical hardware



Building a Secured Multi-Tenant AI Factory

- GPU servers, Storage, Switches, SuperNICs, DPUs, scheduler, Compute & Network Orchestration, GPU orchestration, Security



Without Simulation

24x faster AI factory validation with a Logical Digital Twin



With Simulation

Building a lab	3 days
Software bring-up	0 days
Multi-tenancy	1 week

DSX Air AI factory simulation enables additional flexibility for:

- ✓ Testing at thousand-switch scale
- ✓ Validating alternative solutions
- ✓ Use of different GPUs

End-to-end benchmarking, validation, and optimization approach

From Months **to Minutes**

Shift left the Validation

- Scaled inference validation
- Realistic workload emulation
- Production grade responses

Why It Matters

- Validate at hyperscale before deployment
- Eliminate late-stage failures
- Accelerate production readiness



Traditional vs. Digital Twin

Traditional lab	Digital Twin
128 GPUs	**1K**+ GPUs
80K tokens/sec	**100K**+ tokens/sec
30 days setup	30 min setup

Twin → Production Validation

- Tenant isolation and segmentation
- Network and routing validation
- Inference and capacity emulation
- Security and prompt hardening

Operational Impact

- 60-70% faster validation
- $M+ risk avoided
- 3–6 months faster go-live

Validates AI Infrastructure before you deploying a single GPU

GPU/DPU/NIC | Simulation | Inference Emulation | Kubernetes | Security | Network

End-to-end benchmarking, validation, and optimization approach

From Months to hours



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Traditional vs. Digital Twin

Traditional lab	Digital Twin
128 GPUs	**1K+** GPUs
80K tokens/sec	**100K+** tokens/sec
30 days setup	10 min setup

Operational Impact

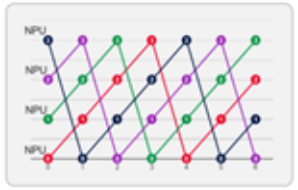
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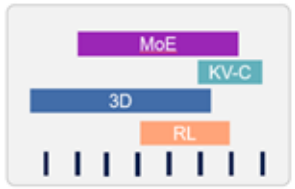
Network Infrastructure - System Lifecycle & Validation Use cases

Collective Benchmarks



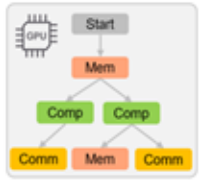
Network Utilization

Mix of Collectives

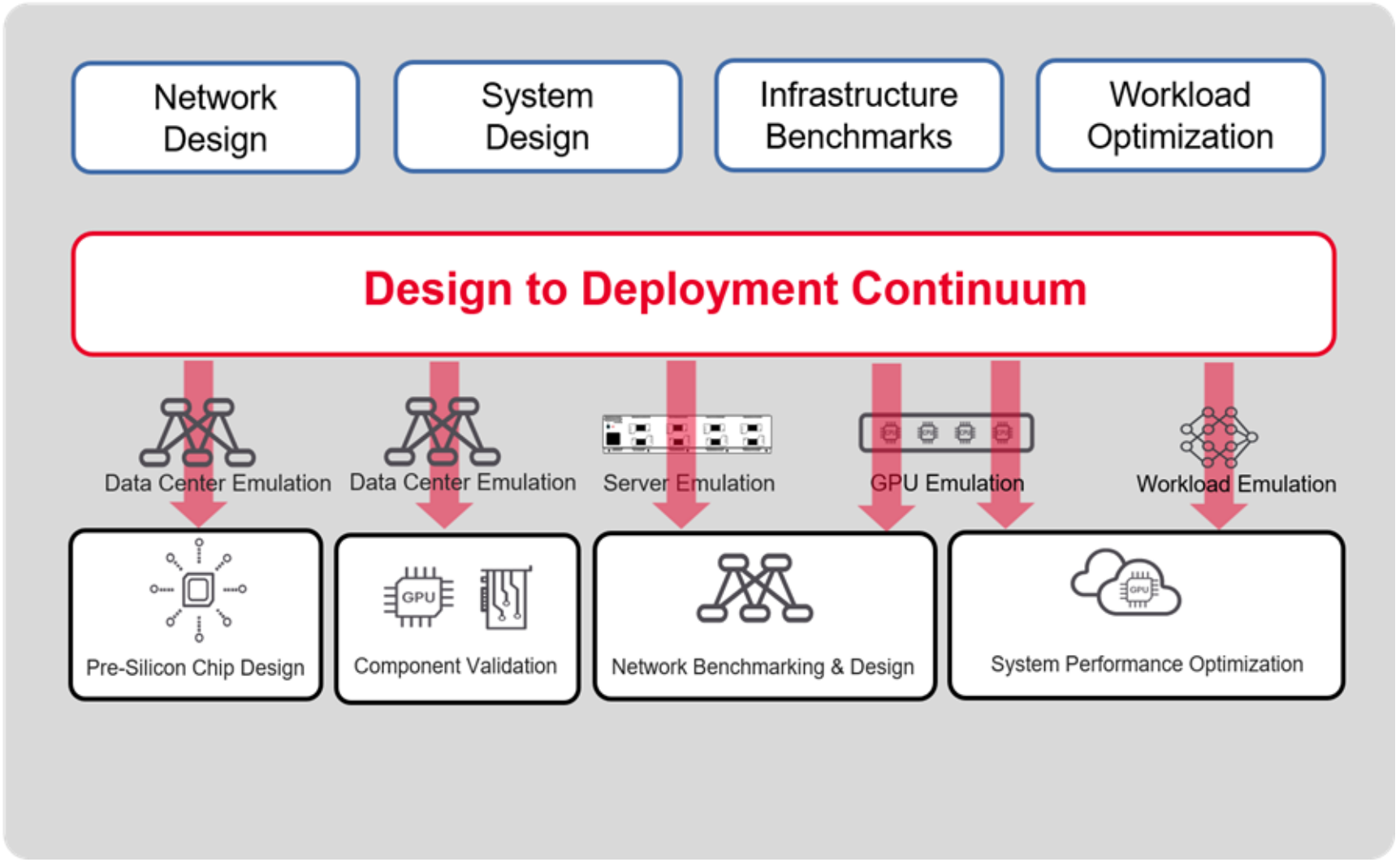


Fabric Validation

Workload Emulation



Training Time



End-to-end benchmarking, validation, and optimization approach

Prompts Determine the resource usage in Inference

Legal Inference

Some legal workloads are dominated by context — massive prompt sizes and long attention windows.

Healthcare Inference

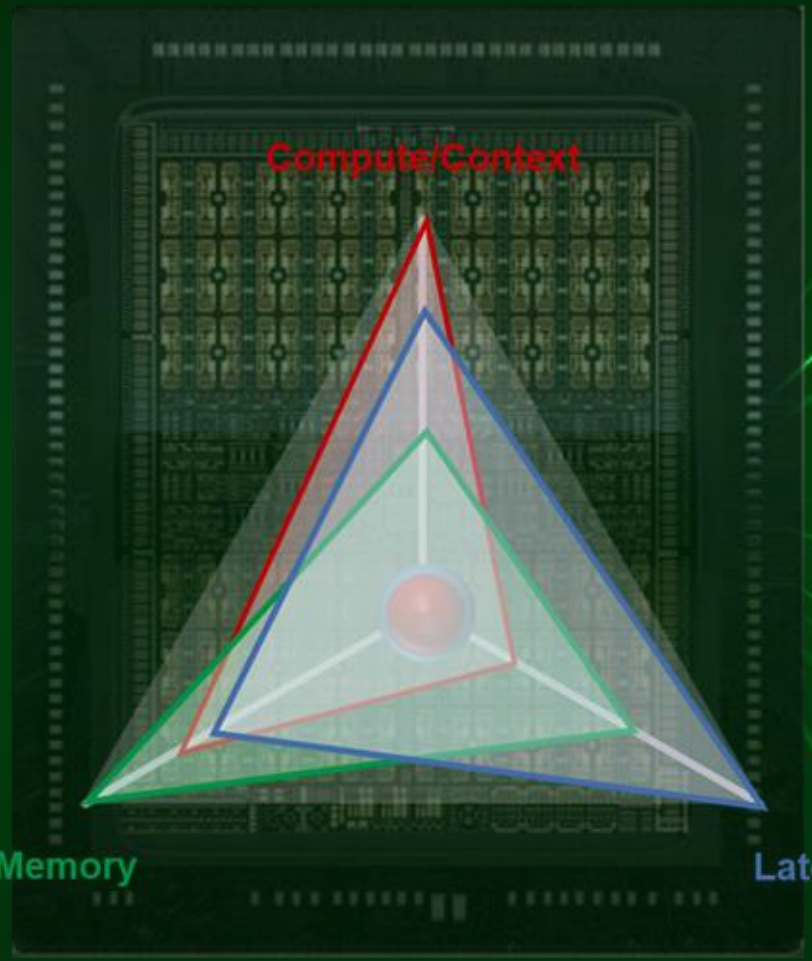
Healthcare inference stresses memory — large images, complex patterns, and sustained throughput.

Financial Inference

Finance cares less about context depth and more about speed — microseconds matter.

Inference is a multi-dimensional problem

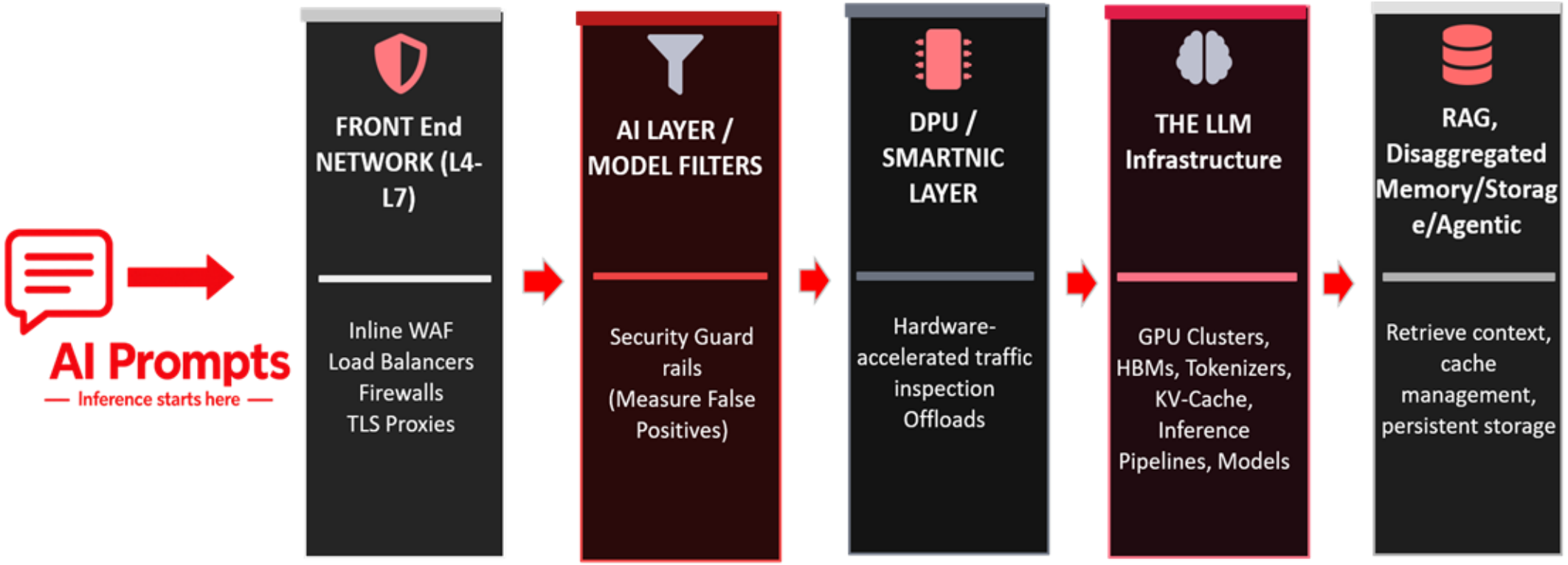
Inference performance isn't about one metric — it's a balance across Engine, Model, Compute, Memory, Storage, Networking, Security etc.



ONE SIZE DOES NOT FIT ALL

Inference is driving workload-specific domain accelerators, custom ASICs, SmartNICs

The Inference Systems – A rich ecosystem of technologies



System Performance and Fine Tuning



Simulate

- Workloads
- Users
- Customization

Validate

- Inference Performance
- Latency characteristics
- GPU / Memory / Power Capabilities

Analyze

- Ingest Inference Telemetry
- Accurately Pinpoint Bottlenecks(Memory/GPU /Network)



Inference Stack Under Test

- PROVIDERS
- WORKFLOWS Model/Stages/...
- ENGINE
- HARDWARE Compute/Memory/Net
- ... Power/Cost/Security

Simulation technologies

- Node Simulation (Logical digital twin) – Higher fidelity
  - Authentic first layer of software stack and APIs



- Cluster / Solution based
  - Cluster simulation
  - Performance simulation
    - Power simulation for a cluster
    - Memory simulation for a cluster
    - Network and compete simulation – modeling

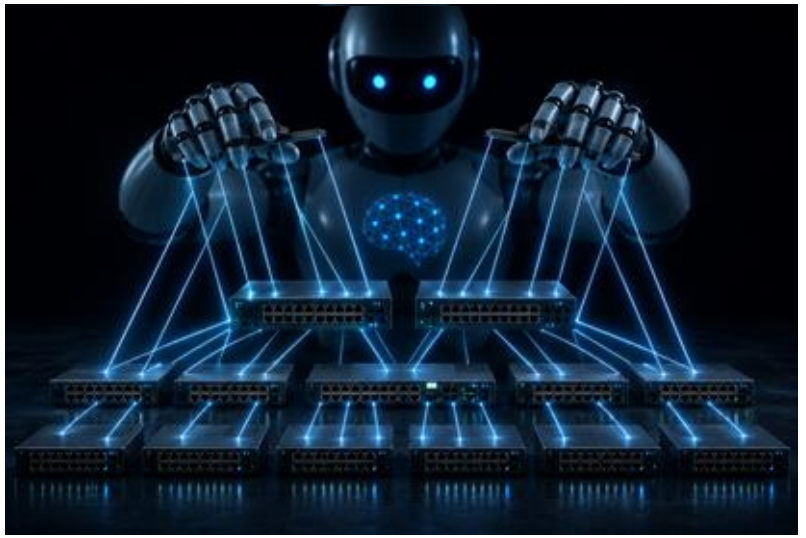


Performance simulation tells you the impact. The Logical Digital Twin tells you what is actually happening.

AI Simulation Technologies

- Agentic AI
  - Against the digital twin – to perform an agentic operation , for example:
    - To validate the digital twin
    - Reporting
    - Security checks and more
  - Embedded as a GPU in the digital twin

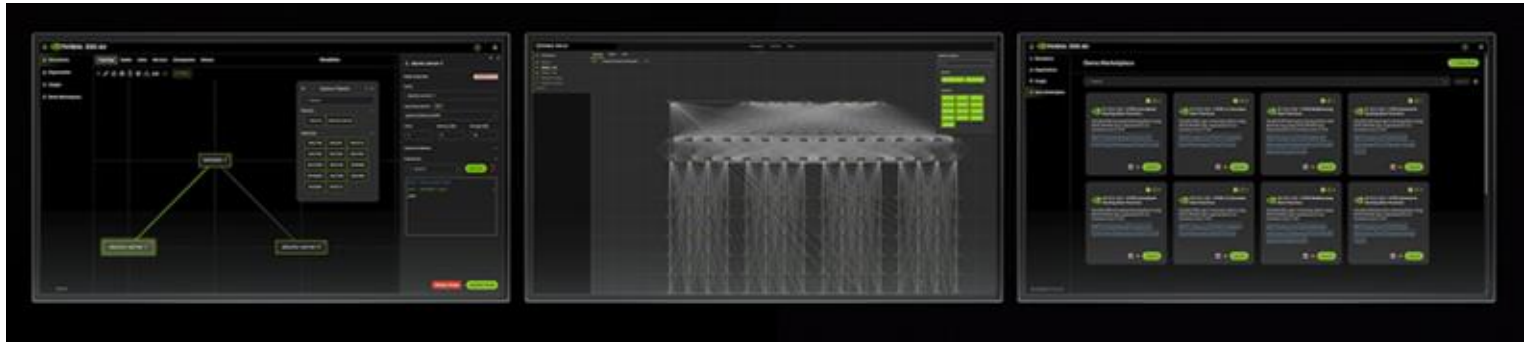
  - What if the switch was actually an AI agent? – In research
    - An agent that behaves like a switch or an entire network ?



Why Multiple Simulation Technologies Must Coexist

Approach	Fidelity	Scale	Best For
Node simulation	High	Large	Integration, Operation and Validation
Solution/Cluster simulation	Medium	Huge	Planning
Performance simulation	High	Large	Validation
AI Agents as simulators	Adaptive	Massive	Operations

NVIDIA DSX AIR & Keysight



- https://www.nvidia.com/en-us/networking/ethernet-switching/air/



KAI Inference Builder



KAI Data Center Builder

Call to Action

- Participate in the OCP Digital Twin Initiative
- Join the DSX AIR community
  - Nvidia offers an EVAL package , contribute your AI Factory simulation and innovate with Nvidia https://docs.nvidia.com/networking-ethernet-software/nvidia-air-v2/Account-Setup/
- Where to find additional information (URL links)
  - Watch GTC 2026 DSX AIR 6 min demo video: https://www.youtube.com/watch?v=kFyud_PpMwA
  - Read the blog: https://blogs.nvidia.com/blog/dsx-air-simulation-ai-factories/
- Validate your AI Factory Simulation
  - Use KAI Inference Builder Lab in DSX AIR
  - Link (TBD)

Thank You!

